



ARTIFICIAL INTELLIGENCE WITH MACHINE LEARNING

COURSE DESCRIPTION

Introduction to Artificial Intelligence

- Introduction to Intelligent Agents
- Problem Formulation
- Informed Search Strategies
- Uniformed Search Strategies
- Constraint Satisfaction Solving problems by Searching
- State Space Formulation
- Depth First and Breadth First Search
- Iterative Deepening
- Heuristics
- Game Playing
- Expert Systems
- Natural Language Processing
- Inductive Learning

Introduction to Machine Learning

- Types of Machine Learning
- Goals and Applications of Machine Learning
- Tools for Machine Learning
 - i. Python for Data Analysis – NumPy and Pandas
 - ii. Python for Data Visualization – Matplotlib, Seaborn and Pandas built-in Data Visualization
 - iii. Image Processing : OpenCV
- **Supervised Learning :**
 - K-Nearest Neighbors
 - K-Means
 - Support Vector Machine Classifier
 - Naive Bayes Classifier
 - Regression Learning
 - Logistic Regression
 - Decision Trees
 - Boosting and Bagging

- **Unsupervised Learning :**
 - Clustering
 - K-means
 - EM Algorithm
 - Mixture of Gaussians
- **Factor Analysis**
 - PCA (Principal Components Analysis)
 - ICA (Independent Components Analysis)
 - Latent Semantic Indexing
 - Markov Models Hidden Markov Models (HMMs)
- **Reinforcement Learning**
 - MDPs. Bellman Equations
 - Linear Quadratic Regulation (LQR)
 - Q-Learning
- **Deep Learning**
 - Introduction to Neural Networks
 - Working on different Inspired Neural Networks (ANN, CNN & RNN/LSTM/GRU)
 - Transfer Learning
- **Projects**
 - Data Prediction using different Machine Learning Algorithms
 - Image Classification
 - Object Detection

Pre-requisite

Course Duration: 40 Hours

Course Type: Certificate

Eligibility Criteria: Knowledge of basic computer programming fundamentals, consistently meeting project deadlines and active participation is all we need.

Minimum System Configuration Required:

- Intel i3 Processor
- 4GB RAM
- 2GB Graphic Card
- 500GB/1 TB HDD
- High Speed Internet (WI-FI)